

Improved low-entropy masking scheme for LED with mitigation against correlation-enhanced collision attacks

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Abstract—Recent lightweight cryptographic algorithms are vulnerable to side channel attacks (SCA), requiring a careful design of implementations. In this paper, we carefully investigate the “Low-Entropy Masking Scheme” (LEMS) on the block cipher LED in the context of resource-constrained environments. We propose an improved LEMS called “iM-LED”, so as to satisfy two different yet unified design goals: to maintain the entropy of the masking at one bit, meanwhile, to enhance first-order SCA-resistance against advanced SCAs such as correlation-enhanced collision attack (CCA). The security of the implementation has been evaluated based on the SASEBO-W board, which proves that iM-LED has boosted its mitigation against SCA with limited overheads.

Keywords-lightweight block cipher, LEMS, LED, SCA.

I. INTRODUCTION

In the past few years, the highly resource-constrained devices are widely used in typical applications of Internet of Things (IoT) and Cyber Physical Systems (CPS), which are designed to be low-cost [1] meanwhile raising serious privacy concerns. Consequently, certain cryptographic primitives should be employed to deal with sensitive applications. If not considering the overheads of resources, standard algorithms such as Advanced Encryption Standard (AES) are well suited. However, the smallest implementation of AES requires 2400 gate equivalents (GE) [2], which exceeds the capacity of most constrained devices. As a result, many lightweight block ciphers have been designed, such as PRESENT [3], LED [4], etc.

Side channel-attack (SCA) is a type of new attack which explores the side channel leakages such as power, timing, and electromagnetism to extract the secret key. Among various SCAs, power analysis attacks (PA) exploit power consumption of devices rather than mathematical properties of cryptographic algorithms in a non-invasive manner, which include simple power analysis (SPA), differential power analysis (DPA), correlation power analysis (CPA), etc [5].

Lightweight ciphers do resist classical cryptanalysis, yet, they are more vulnerable to SCAs than standard cryptography (e.g. AES) due to the relatively compact and simple structure. So SCA countermeasures for lightweight ciphers

should be provided in some sensitive applications to boost their security. The countermeasures to resist SCA can be categorized into two distinct groups: masking and hiding [5]. Hiding technology is to remove the data dependency of the power consumption. Masking technology is to randomize the processed intermediate values by mixing with a random yet unknown value. Due to the fact that masking is transparent to the adversary and does not require any unique backend balance, we concentrate on this option in the rest of the paper. However, when coping with resource-constrained devices, complicated countermeasures might be difficult to deploy because of the non-negligible cost in terms of area, performance or both [1]. Thus some compact and low-overhead countermeasures should be taken into consideration for lightweight cryptography.

In general, the costs imposed by masking schemes are quite high in terms of memory and computation time. Therefore the low-entropy masking scheme (LEMS) is proposed so as to minimize its overheads. The “Rotating SBoxes Masking” (RSM) scheme is one kind of LEMS against SCAs on AES [6]. RSM gains its superiority by taking no time penalty, costing reasonable area, applying only 4 bit masking entropy and still keeping secure against univariate first- and second-order attacks. Bhasin et al. [1] have generalized the LEMS concept of RSM and presented an approach on PRESENT with only one-bit entropy, *i.e.*, the approach employs just two different masks and can achieve a provable first-degree security. However due to the appearance of side channel analysis such as collision correlation analysis (CCA) [8], the vulnerability of one-bit rotating SBoxes masking scheme is turning into a realistic threat when facing newly advanced side-channel attacks.

In this paper, we first implement the low-entropy RSM-based masking scheme and verify its vulnerability against collision correlation analysis. In order to boost the security with limited overheads, we stick to the one bit entropy solution and propose an enhanced masking scheme with some optimizations. All of our designs are illustrated with the application to the block cipher LED (Light Encryption Device) which is newer than PRESENT and designed with

a more compact structure. However our method is generic and should also work with other lightweight block ciphers including PRESENT.

The paper is organized as follows. Section II introduces some technical background. Section III elaborates our implementation of LED with one-bit entropy RSM scheme and evaluates it with CPA and CCA. Section IV describes our improved scheme of iM-LED with further verifications. Section V includes some discussions and the conclusion.

II. BACKGROUND

A. The LED Block Cipher

LED is an AES-like SPN-type symmetric cipher, whose block size is 64-bit. It has two primary versions involving 64-bit and 128-bit keys: LED-64 and LED-128 [4]. The cipher state is conceptually arranged in a 4×4 vector. In the rest of this paper, we focus on LED-64 for lightweight applications and abbreviate it as LED.

The encryption procedure of LED is made up of two primary operations: addRoundKey and step. Note that there is no key scheduling in LED. The plaintext P is XORed with the secret key K in the addRoundKey operation (denoted as AK) and then goes through the step operation eight times. Then the value from the last step operation is XORed with K again and becomes the ciphertext. Here, the step operation consists of four rounds of encryption for the state and the state is XORed with K only between steps. Each round function is made up of AddConstants (AC), SubCells (SC), ShiftRows (SR) and MixColumnSerial (MC) operations, which resembles the structure of AES.

B. CPA and TVLA

Correlation power analysis (CPA) and Test-vector-leakage-assessment (TVLA) are two common types of methods for SCA evaluations. CPA was proposed by Clavier et al. [7]. The hypothetical power consumption model of the CPA was set to the Hamming weight model. In realistic cases with a set of power traces T_i ($1 \leq i \leq N$, where N is the total number of traces), the correlation can be computed as ρ , the Pearson Correlation Coefficient, as in Eq.(1). σ_X is the standard deviation of X . $H_{i,j}$ is the hypothetical power consumption model which can be calculated from the j -th key hypothesis and the corresponding plaintext.

$$\rho_{TH} = \frac{\text{cov}(TH)}{\sigma_T \sigma_H} = \frac{N \sum T_i H_{i,j} - \sum T_i \sum H_{i,j}}{\sqrt{N \sum T_i^2 - (\sum T_i)^2} \sqrt{N \sum H_{i,j}^2 - (\sum H_{i,j})^2}} \quad (1)$$

TVLA [11] has been standardized as a metric to assess the potential side-channel leakage, which only relies on the measured traces. In principle, two groups of traces are required for TVLA computation: the first trace group A corresponds to a fixed key and fixed plaintexts, serving as the reference; the second trace group B corresponds to random

plaintexts. Denote (X_A, S_A) and (X_B, S_B) - the mean and the standard deviation of the traces in group A and B , respectively. Let N_A and N_B be the size of A and B . Then the t -statistic trace T (over the same time instants) is computed by Eq.(2). If the general selected secure leakage range is between $+4.5$ and -4.5 , the acceptance or rejection confidence level can easily reach at 99.999%.

$$T = \frac{X_A - X_B}{\sqrt{S_A^2/N_A + S_B^2/N_B}} \quad (2)$$

C. RSM

A boolean masking scheme called RSM, is firstly proposed to protect the non-linear operation of AES in [6], which is robust against univariate first-order and zero-offset second-order SCA. The RSM scheme is employed in DPA contest V4. By carefully selecting the mask sets, RSM becomes a variant realization of LEMS, which keeps the same level of timing performance as an unprotected design with only a modest area overhead. The efficiency is mainly attributed to the starting rotation offset with only 4-bit entropy while in other masking schemes the number of random bits is equal to (or multiples of) their state size. The RSM scheme consists of the following two parts:

MaskedSubBytes: Firstly 16 masked SBoxes are pre-computed once for all. Then after randomly choosing a starting rotation offset j for each encryption, the new SBox operation (S') of masked state X' (where $X' = X \oplus M_j$) can be expressed by the following equation: $S'_j = S(X' \oplus M_j) \oplus M_{j+1 \bmod 16}, \forall j \in \{0 - 15\}$. M_j is the j -th mask in the mask set.

MaskCompensation: Due to the non-linear operations, the state is XORed with a corresponding compensation, $CSM_j = (M_j) \oplus SR(MC(M_j))$, to make sure that the state in the next round is masked by M_{j+1} . CSM stands for the compensation mask. The SBoxes are rotated by one position in a fixed direction. The mask compensation of final round is specially computed due to the lack of MC operation.

III. IMPLEMENTATION AND EVALUATION OF M-LED

In this section, we give the details of our implementation of masked-LED algorithm, and explore two attacks. We investigate more on the so-called correlation-enhanced collision analysis (CCA) which is considered as a more advanced SCA.

A. Implementation of one-bit RSM for LED

First, we implement the one-bit low-entropy RSM scheme for LED to evaluate its SCA resistance, realizing a primary masked-LED (M-LED). In Fig.1, the original SBox S is replaced by modified SBoxes ($S'_{m_i}, i \in \{0, 1\}$) according to the chosen 4-bit masks m_i , which should satisfy:

$$S'_{m_i}(x') = S(x' \oplus m_i) \oplus m_{i+1 \bmod 2} \quad (3)$$

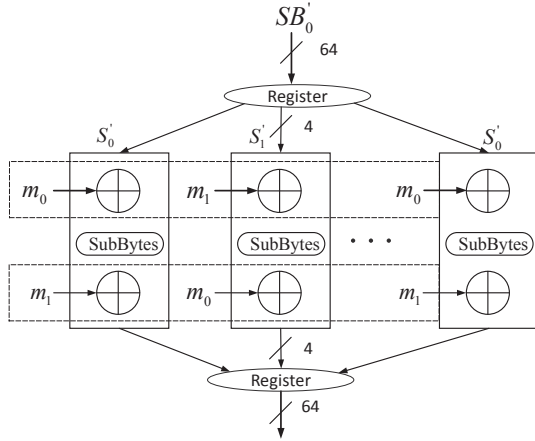


Figure 1. Low-entropy (one-bit) RSM-based SBox.

Second, a random offset j is generated by a pseudo random number generator (PRNG) to choose the corresponding SBox, *i.e.*, to choose the corresponding mask for each nibble of state. So the 64-bit masks M_0 and M_1 can be expressed by $\{m_0, m_1, \dots, m_0, m_1\}$ and $\{m_1, m_0, \dots, m_1, m_0\}$ respectively. We make $(m_0, m_1) = (0x5, 0xA)$ in the M-LED scheme just the same as [1]. So in the SC operation, for every masked state $x' = x \oplus m_i$, the output of SBox is masked by $m_{(i+1) \bmod 2}$, meaning that the SBoxes are rotated by one nibble.

Third, masking the SR , MC and AK functions only requires a simple XOR operation due to their linear properties. We use the mask compensation set CSM_i ($i = 0$ or 1) to make the state again masked by $M_{(i+1) \bmod 2}$ in preparation for the next round. Note that this procedure is a little different from masked-PRESENT in [1]. Here, CSM_i is computed as follows:

$$CSM_i = (M_i) \oplus SR(MC(M_i)), \forall i \in \{0, 1\} \quad (4)$$

The pseudo code of our initial implementation of the one-bit entropy masking scheme for LED is shown as Alg.1.

B. Experiment Platform

We employ an 8-bit microcontroller Atmega163 embedded in a smartcard which is read using a simple reader interface mounted on the evaluation board SASEBO-W and controlled by a Xilinx Spartan-6 FPGA. The power traces corresponding to random plaintexts are acquired using an Agilent DSO9404A oscilloscope with a probe at a sampling rate of 100 MSa/s for a duration of $500 \mu\text{s}$. A waveform of one M-LED encryption includes 2.2×10^5 sampling points. The keys employed in our experiment are priori data $\{0x0, 0x1, \dots, 0xF\}$. Here, we denote MTD for the minimum number of traces to disclose part or all of secret keys. Our simple experiment of CPA attack on the unprotected LED shows that the MTD to reveal all the 16 nibble keys is about 30.

Algorithm 1: M-LED

Input: $m_0 : \text{Mask} \in [3, 5, 6, 9, a, c]$; n_{trace} : number of encryption; P : Array of plaintext
Output: C : Array of ciphertext

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1 Define:  $m_1 \in \mathbb{F}_2^4$ , and  $m_1 = \neg m_0$ ;  $\text{offset} = j$ ;
    $M_0 = \{m_0, m_1, \dots, m_1\}$ ;
    $M_1 = \{m_1, m_0, \dots, m_0\}$ 
2 Compute: Mask SBoxes  $S'_{m_0}$  and  $S'_{m_1}$ ;
   //  $S'_{m_i}(x') = S(x' \oplus m_i) \oplus m_{i+1 \bmod 2}$ 
   Mask Compensation  $CSM_i, \forall i \in \{0-1\}$ 
3 for  $i = 1$  to  $n_{\text{trace}}$  do
4    $j = \text{rand}() \% 2$ ;
5    $\text{state} \leftarrow P[i]; \text{key} \leftarrow K[i]$ ;
6   LEDRound_mask ( $j, \text{state}, \text{keys}$ )
7   if  $j = 1$  then
8      $\text{offset}_{\text{index}} = 0xAAAA, \text{state} \leftarrow M_1$ 
9   else
10     $\text{offset}_{\text{index}} = 0x5555, \text{state} \leftarrow M_0$ 
11    AddKeys( $\text{state}, \text{keys}$ );
12    for  $\text{round} = 1$  to  $32$  do
13      AddConstants( $\text{state}$ );
14      SubCell( $\text{state}$ );
15      ShiftRow( $\text{state}$ );
16      MixColumn( $\text{state}$ );
17      Maskcompensation( $j, \text{state}$ );
18       $j = (j + 1) \% 2$ ;
19      if  $j = 1$  then
20         $\text{offset}_{\text{index}} = 0xAAAA, \text{state} \leftarrow M_1$ 
21      else
22         $\text{offset}_{\text{index}} = 0x5555, \text{state} \leftarrow M_0$ 
23      AddKeys( $\text{state}, \text{keys}$ );
24    return state

```

C. Evaluation of M-LED against CPA

The MTD of M-LED to reveal part of the secret keys is shown in Fig.2. As to the adversary, performing a successful CPA attack with limited power traces is quite difficult. Specially in this case with up to 10^4 traces, a standard CPA reveals at most 3 of the total 16 nibble keys. Compared with the unprotected LED requiring only 30 traces to extract all 16 key nibbles, the expecting mitigation of M-LED against CPA has been achieved. The difficulty for the adversary to launch a standard CPA is significantly increased.

D. Evaluation of M-LED against CCA

The correlation-enhanced collision attack (CCA) [8] in nature belongs to the internal template-based SCA. In [8], a collision occurs when two SBoxes at the byte position $j_1 = a$ and $j_2 = b$ collide, with the assumption that $P_a \oplus K_a = P_b \oplus K_b$, where P_j and K_j denotes the j -th byte in the plaintext and key, respectively. Consequently, CCA requires neither the access to a copy of the target device with a known key nor the so-called profiling phase. As long as the means of the power consumption traces for certain intermediate values are distinguishable, the CCA attack can succeed. Therefore, CCA works well against some masked implementations, which still have first-order leakage.

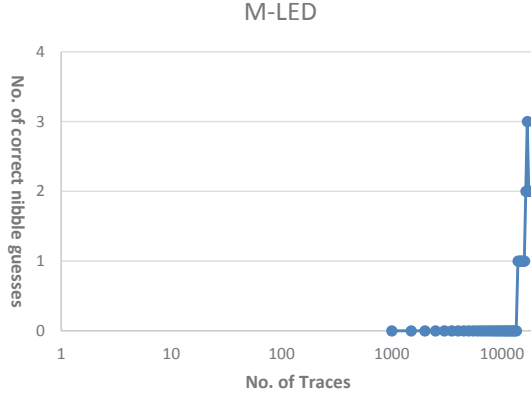


Figure 2. No. of correct nibble guesses v.s. No. of traces for M-LED

In this context, we employ this attack to exploit the XOR-difference between two key nibbles in M-LED. Note that two adjacent SBoxes are masked by m_i and $m_{i+1 \bmod 2}$ (denoted as m_{i+1} for short) respectively and $m_i \oplus m_{i+1} = 0xF, i \in \{0-15\}$. We assume $k_{guess} = k_i \oplus k_{i+1}$. Then based on the collision $x_i = x_{i+1} \oplus k_{guess}$ and the bijective characteristic of SBox, we can infer that:

$$S'(x'_i \oplus k_i) = S'(x'_{i+1} \oplus k_{i+1}) \oplus 0xF, \forall i \in \{0-15\} \quad (5)$$

Hence, we can reveal the k_{guess} based on this collision using corresponding power leakages. Furthermore, CCA can exploit most of first-order leakages without the accurate knowledge of the hypothetical power model. In practice, the power models used in CPA are not that precise as in reality due to the complex static power (especially in shrinking technology environments) and noise. Since CCA utilizes two different segments of the exact same power trace with higher correlation, CCA is more powerful when compared to CPA.

In this experiment, we collect 3×10^4 traces corresponding to randomly chosen plaintexts and offsets. The CCA attack calculates the correlation between the power dissipations of two instances of SBox operations. The results show that with at least about 1.8×10^4 traces, parts of the XOR-keys recovery can be successful. For example, we applied the keys $\{0x0, 0x1, \dots, 0xF\}$ in the attack and the CCA results of k_{guess} ($k_2 \oplus k_3$ and $k_3 \oplus k_4$) were successful as shown in Fig.3. With about 2.5×10^4 power traces, all the XOR-keys of M-LED can be extracted, which reveals the vulnerability of M-LED under the CCA attacks.

In practice, the guessing results with relatively high correlation are also considered as alternatives of real key recovery, such as the 8-th, 9-th and 13-th hypothesis for $\{k_2 \oplus k_3$ in Fig.3a. Then we apply “mutual proof verification” method to get the real value[10]. Since parts of the keys bits (*i.e.*, the exclusive XOR), are revealed, the rest of unknown key bits can be easily recovered by exhaustive search.

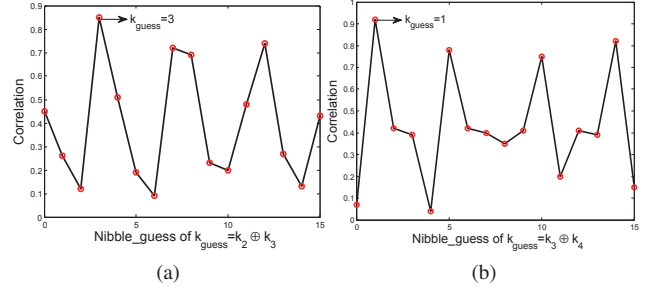


Figure 3. Examples of CCA attack results against M-LED. The corresponding key guesses have the maximal correlation coefficients.

IV. IMPROVEMENT AND OPTIMIZATIONS OF IM-LED

Since the M-LED algorithm only employs as low as one-bit entropy randomness, it may still be vulnerable to SPA or high-order CPA. Trying to thwart all attacks at once is not an easy task, especially in resource-constrained hardware implementations. So in the following, we try to boost the security level of the M-LED implementation by the following methods at reasonable overhead. Note that we stick to the one-bit RSM which holds the property of lightweight, and make only slight modifications to mainly cope with the mitigation against CCA at the first priority.

- **Randomize masks.** Note that only two fixed masks (e.g. $0xA$ and $0x5$) are employed in the M-LED, so the masks corresponding to the same operation can be aligned in all power traces statistically, which increases the success rate of SCA. Choosing a balanced mask is of great importance for the purpose of security under the condition that $m_0 \oplus m_1 = 0xF$ and $HW[m_0] = HW[m_1]$, so $(m_0, m_1) \in \{0x3, 0x5, 0x6, 0x9, 0xA, 0xC\}$. Then we can randomly choose the masks from the mask sets to shuffle the intermediate data at the same location of the algorithm. As a result, the entropy for each plaintext is still 1-bit, yet the difficulty for successful CPA attack is getting increased.
- **Disrupt the SC order.** In the general process, the 16-nibble plaintexts masked with m_0 and m_1 are handled alternatively in rotation, which leads to the leakage in CCA attack. And the SubCell operations are always the targets of SCA. So we apply the method of disrupting the SC order to conceal the above leakage. This strategy can be realized by applying a random number as the index of the first SBox to be handled and then operating other consecutive SBoxes in turns, which makes it difficult to correlate power traces in the time domain. While for resource-constrained environments, we can just disrupt the SC order of the first and the last round in M-LED. For other rounds, the SBoxes stay unchanged.
- **Random inversion.** Y. Lu et al. have proven that by

inverting the inputs to the AddRoundkey operation randomly, the modified AES can resist traditional first-order DPA [9]. The similar technique can be embedded in our design. Whether to employ the original or inversed inputs can be based on a new random offset.

Algorithm 2: Randomize masks

Input: $m_0 : \text{Mask} \in [3, 5, 6, 9, a, c]$;
 flag_1 ; // random mask index

1 **Define:** $m_1 \in \mathbb{F}_2^4$, and $m_1 = \neg m_0$;
 $M_0 = \{m_0, m_1, \dots, m_1\}$; $M_1 = \{m_1, m_0, \dots, m_0\}$

2 **Compute**(m_0):

3 Mask SBoxes S'_{m_0} and S'_{m_1} ;
// $S'_{m_i}(x') = S(x' \oplus m_i) \oplus m_{i+1 \bmod 2}$

4 Mask Compensation $CSM_i, \forall i \in \{0-1\}$

5 **for** $i = 1$ **to** n_{trace} **do**

6 $\text{flag_1} = \text{rand}() \% 3$;

7 **switch** flag_1 **do**

8 **case** 0

$m_0 = 3$; **Compute**(m_0); **break**;

9 **case** 1

$m_0 = 5$; **Compute**(m_0); **break**;

10 **otherwise**

$m_0 = 6$; **Compute**(m_0); **break**;

Algorithm 3: Disrupt the SC order

1 **Input:** $\text{count}=0$;
 flag_2 ; // random SBox index

2 **for** $i = 1$ **to** n_{trace} **do**

 // Shuffling

3 **for** $\text{round} = 1$ **to** 32 **do**

4 $\text{flag_2} = \text{rand}() \% 16$;

5 **for** $\text{count} = 1$ **to** 16 **do**

6 $\text{SubCell}(\text{flag_2})$; // Firstly operate the

7 flag_2 SBox ;

$(\text{flag_2}++) \% 16$;

The algorithms of Randomizing masks and Disrupting the SC order are described in Alg.2 and 3, respectively. The overheads of these two algorithms only include two PRNGs and a couple of registers. We have extended the M-LED with above two algorithms, resulting in an improved version called iM-LED. Then we have mounted CPA and CCA attacks to prove its enhanced security in practice.

In terms of CPA attack, we have collected 10^5 power traces with 2.2×10^5 points per trace. Due to the constrained RAM storage, we only attack the first six key nibbles. The MTD of iM-LED to reveal part of the secret keys is shown in Fig.4. Obviously, the right key guesses are well-masked in all hypotheses, and using 10^5 power traces is still unable to reveal all the secret keys. Consequently, we believe that the relatively high-level security of the iM-LED is realized. Furthermore, the CCA attack results of two-nibble pairs ($k_2 \oplus k_3, k_3 \oplus k_4$) prove that the robustness of the iM-LED

against CCA attack is significantly improved, because with up to 6×10^4 power traces we still cannot extract the right XOR of key nibbles.

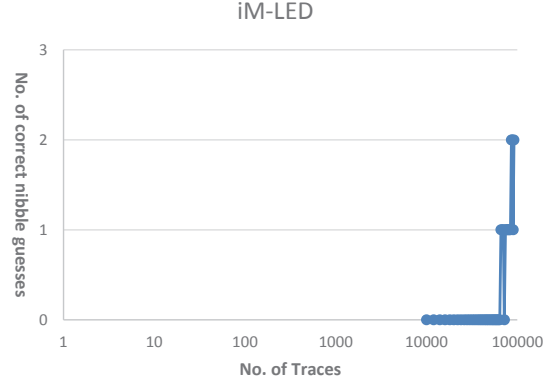


Figure 4. The found key index against nibble number of iM-LED

Besides CPA and CCA methods, we have also computed the TVLA leakages of LED, M-LED and iM-LED implementations for the 5-th nibble plaintext. Here, the number of traces that are used to compute TVLA for all implementations is 5000. Note that various LED implementations take different execution times, which result in different length in time domain. As shown in Fig. 5, the iM-LED (red curve) presents the least leakages when compared with the LED (green curve) and M-LED (black curve), which is the only one smaller than the generic threshold values (± 4.5). Therefore, we can conclude that our iM-LED implementation passes TVLA leakage test.

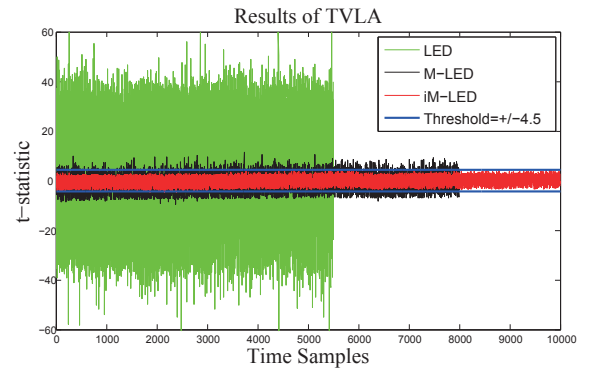


Figure 5. TVLA leakage assessment for various implementations.

In addition, the overhead comparison of various LED implementations is shown in Table I. “text” is the size of code segment which stores the instructions. “data” stands for the size of initialized variables in RAM and FLASH. “bss” is the length of the bss section. According to the simulation results from above sections, we find that the M-LED can improve its security against the original LED with only about 1.5 times of overhead. While, based on M-LED, with

merely about 1.14 times of overhead for tradeoff, iM-LED can further increase its security. So the proposed algorithms are promising to be used in the security-critical but resource-constrained cryptographic applications.

Table I
OVERHEAD OF VARIOUS LED IMPLEMENTATIONS (UNIT: BYTES).

Algorithms	LED	M-LED	iM-LED
text	4076	6254	6906
data	188	232	456
bss	84	114	178
dec	4348	6600	7540

V. DISCUSSION AND CONCLUSION

In this paper, we explore the lightweight countermeasure of LED implementations and investigate its mitigation against CPA and CCA. The reason that we focus on LED is that: it is designed after PRESENT and dedicated for much smaller implementation cost (eg. area (GE) and clock cycles per block) than other block ciphers with similar key and block size, such as PRESENT and KLEIN.

We carefully investigate CCA on LED. One reason is that CCA actually relies on the correlation between two SBox lookups in the same trace, which is considered as more accurate than traditional CPA. The other reason is that CCA explores the relations between the masks in different rounds, which is quite efficient in attacking the countermeasures with simple and linear updates of maskings. We verify CCA on M-LED which takes a straight-forward adoption of RSM from PRESENT to LED.

As a result, some optimizations have been presented to the proposed algorithm with limited overheads, resulting in the proposed iM-LED, whose enhanced security has been evaluated by the corresponding experiments, including CCA and TVLA. Note that we are more interested in mitigating the possible attack such as CCA and demonstrate the efficiency of our countermeasure against correlation based attacks. Since we are not pursuing a full and perfect countermeasure in such a resource-constrained scenario, other advanced attacking strategies and corresponding countermeasures are not investigated. For the purpose of mitigation, we only use a few thousands of traces to demonstrate the feasibility and efficiency, therefore we did not collect millions of traces as traditional SCAs to conduct a full attack.

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